Quantum Transport

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The 1D stationary Schrödinger equation

$$\left[-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial z^2} + U(z)\right]\psi(z) = E\,\psi(z)$$

- $U(z)$ is piecewise-constant (alternating well and barrier layers); m_e is the effective mass under the parabolic-band approximation.
- In each layer the solution is a superposition of progressive and regressive plane waves $A\,e^{+ik_z z} + B\,e^{-ik_z z}$ with $k_z = \sqrt{2m_e(E-U)/\hbar^2}$.

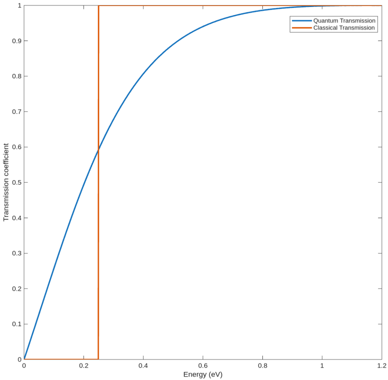
Quantum mechanics $\leftrightarrow$ transmission-line analogy

Quantum mechanics	**Transmission line**
Wavefunction $\psi(z)$	Voltage $V(z)$
Probability current $J(z)$	Active power $\frac{1}{2}\mathrm{Re}\{VI^*\}$
$\sqrt{2m_e(E-U)/\hbar^2}$	Wavenumber k_z
$\sqrt{8(E-U)/m_e}$	Characteristic admittance Y_∞

- Transmittance, LDOS and bound states are obtained from the local reflection coefficients $\overleftarrow{\Gamma}(z)$, $\overrightarrow{\Gamma}(z)$ rather than from transfer matrices.

Computing the transmission coefficient

Singlr barrier



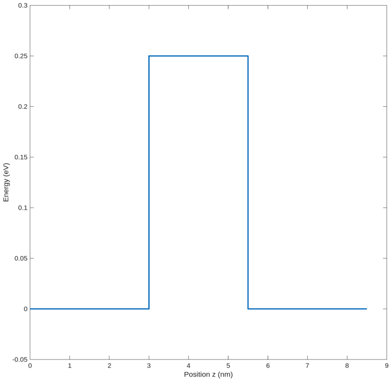


Figure: Quantum transmission coefficient $T(E)$ of a single rectangular barrier compared with the classical step prediction, showing sub-barrier tunnelling and the characteristic over-barrier oscillations.

Computing the transmission coefficient
Thin barrier

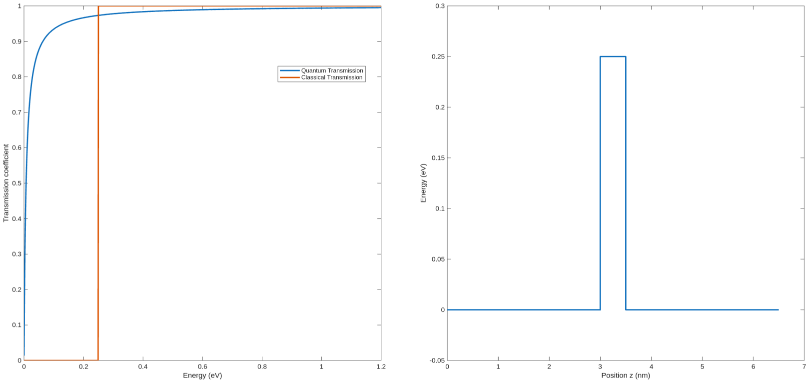


Figure: Transmission coefficient of a thin barrier

Computing the transmission coefficient
Thick barrier

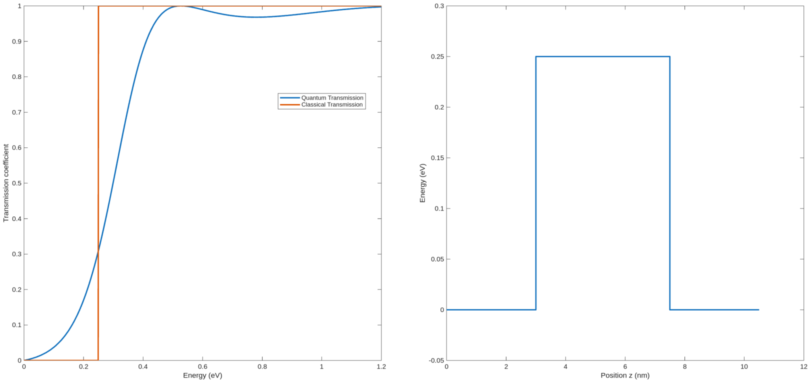


Figure: Transmission coefficient of a thick barrier

Computing the transmission coefficient

Higher barrier

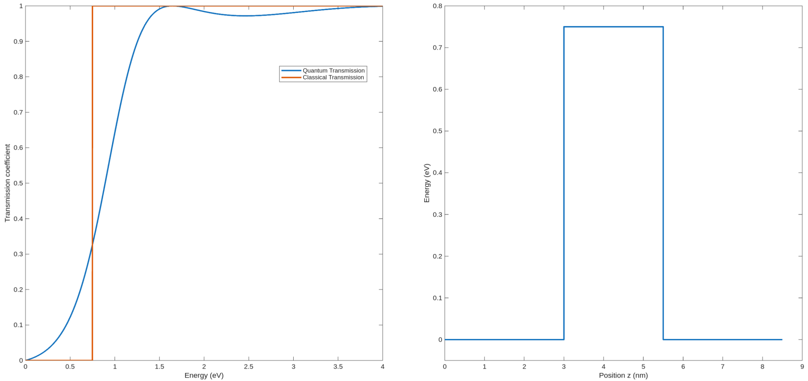


Figure: Transmission coefficient of a higher barrier

Computing the Local Density of States

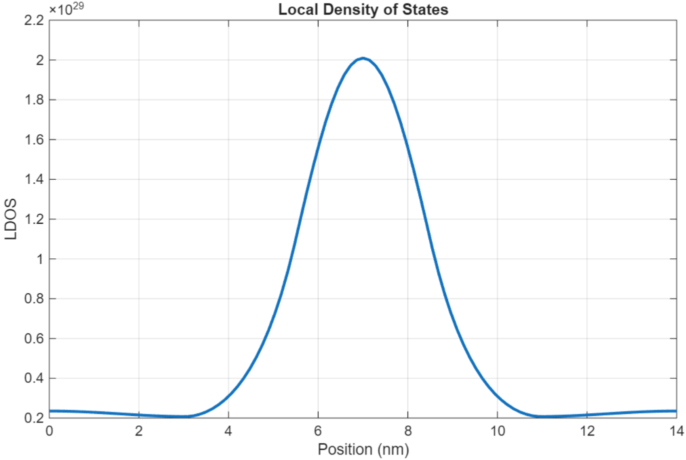


Figure: Local density of states along the growth direction of a double-barrier resonant structure at an energy lying inside the well, evidencing the standing-wave pattern of the quasi-bound state.

Transmittance spectrum
Single barrier

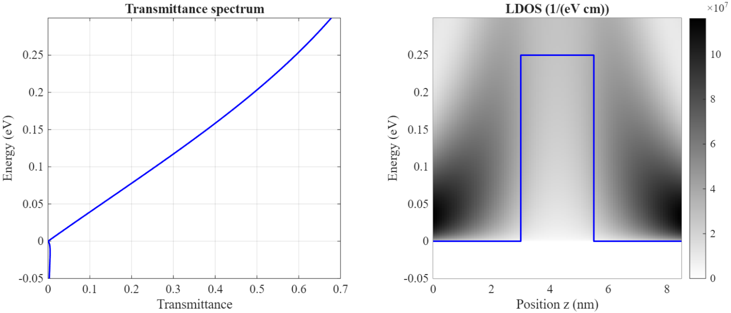


Figure: Transmittance spectrum (left) and energy-resolved LDOS map (right) for a single barrier: smooth tunnelling tail below U_{barr} and Ramsauer-like oscillations above.

Transmittance spectrum
Double barrier

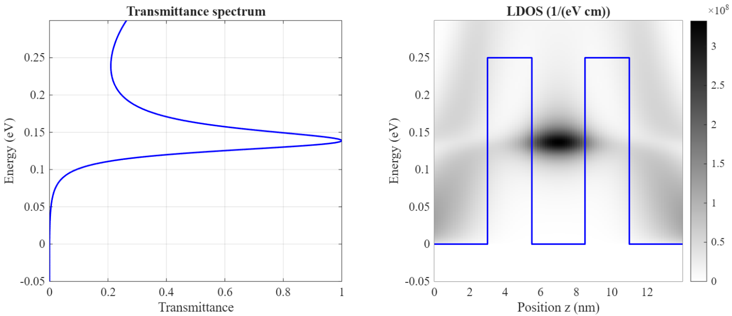


Figure: Transmittance and LDOS for a symmetric double-barrier structure, with the sharp resonant peak corresponding to the quasi-bound state localised in the central well.

Transmittance spectrum
20 barriers

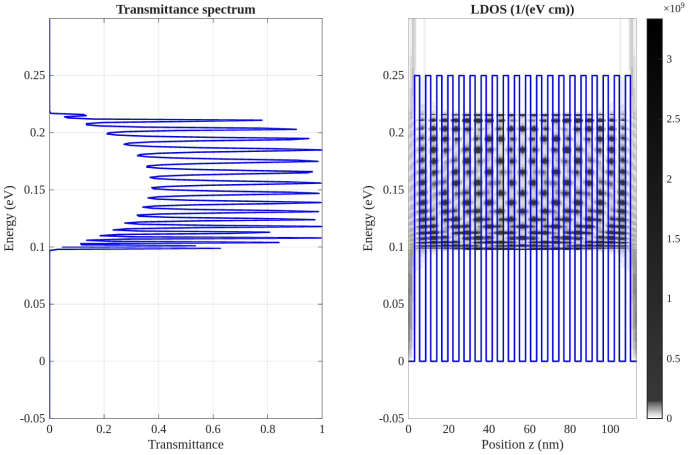


Figure: Transmittance and LDOS for a 20-period superlattice: the discrete resonances of the few-barrier case merge into allowed minibands separated by minigaps.

Computing Bound States

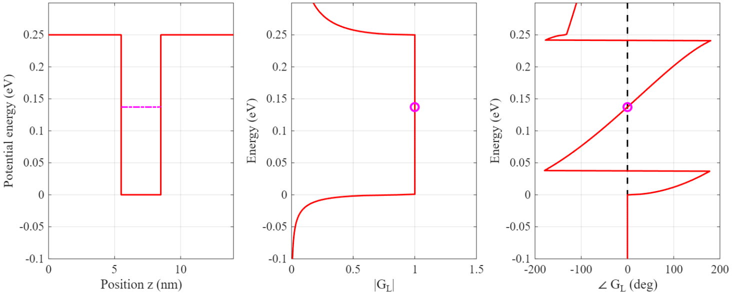


Figure: Bound-state search by the loop-gain method: potential profile (left), $|G_L(E)|$ (centre) and $\angle G_L(E)$ (right); the magenta marker locates the bound-state energy found by a phase-zero crossing.

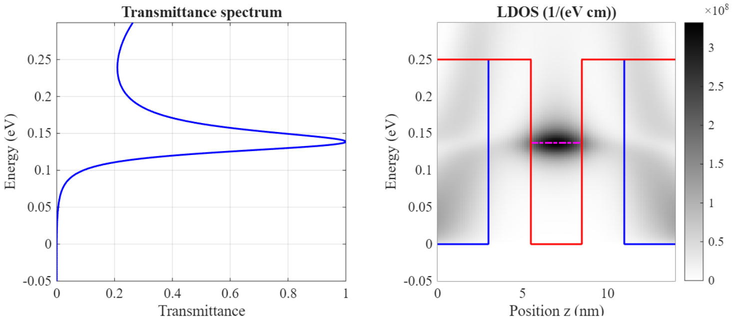


Figure: Comparison of the bound-state energy found by the loop-gain method with the LDOS map

Transmittance spectrum

No electric field

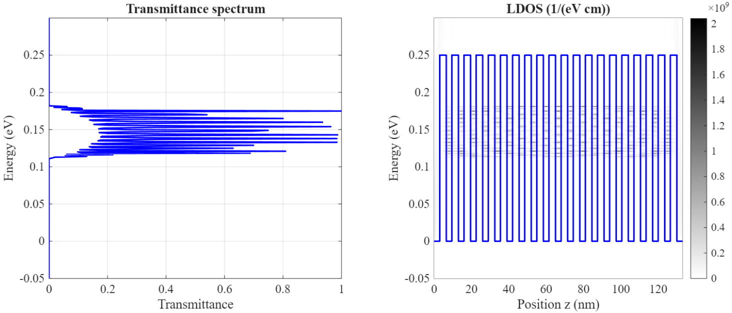


Figure: Transmittance spectrum (left) and LDOS map (right) for an unbiased 20-period superlattice ($t_{\text{barr}} = 2.5$ nm), showing flat minibands extending uniformly across the device.

Transmittance spectrum with Electric field

Voltage applied: 0.025 V

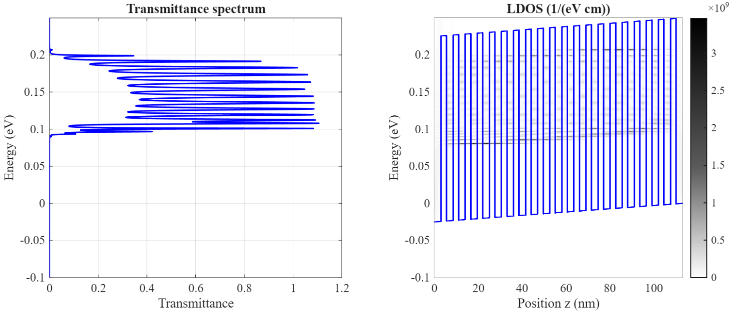


Figure: Same quantities for a 20-period superlattice ($t_{\mathrm{barr}} = 2.5$ nm) under an applied bias of $V = 0.025$ V: the linear potential drop tilts the minibands and breaks the resonant alignment between adjacent wells (note that t_{barr} differs from the unbiased case in the previous slide, so the comparison is qualitative).

Transmittance spectrum with Electric field

Voltage applied: 0.05 V

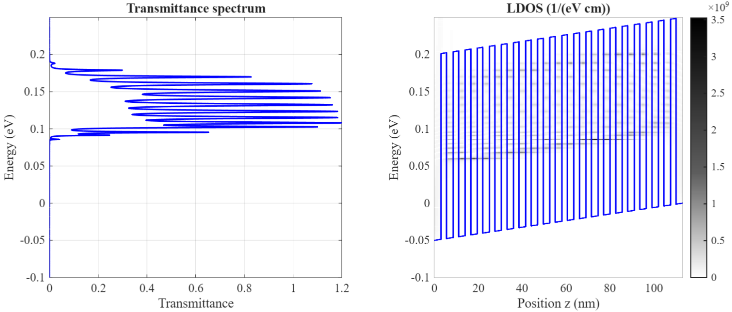


Figure: Same lattice ($t_{\text{barr}} = 2.5$ nm) under an applied bias of $V = 0.05$ V.

Transmittance spectrum with Electric field

Voltage applied: 0.075 V

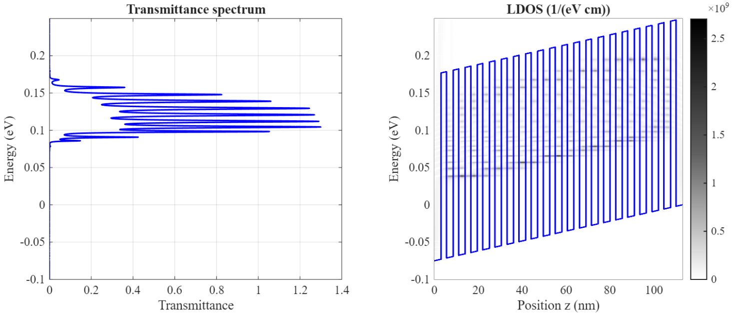


Figure: Same lattice under an applied bias of $V = 0.075$ V.

Carrier Velocity

Varying barrier thickness

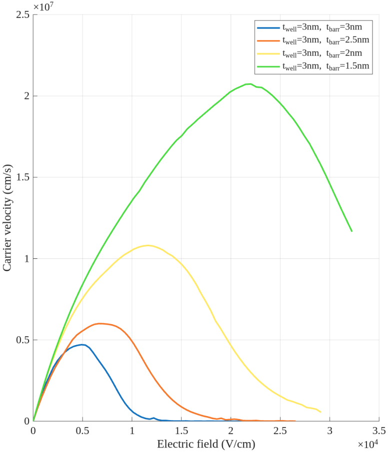


Figure: Carrier velocity as a function of energy for different barrier thicknesses.

Carrier Velocity

Varying well thickness

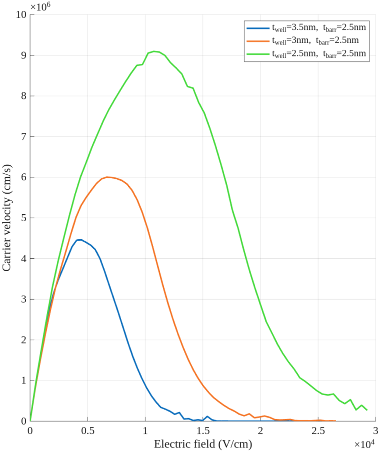


Figure: Carrier velocity as a function of energy for different well thicknesses.